An Unequal Clustering Method Based on Particle Swarm Optimization in Underwater Acoustic Sensor Networks

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***Abstract*—Underwater acoustic sensor networks (UASNs) currently provide an important technical means of underwater communication, but there are difficulties in power updating or power replenishment because the sensor nodes work in an underwater environment. Therefore, energy consumption optimization has become the focus of research on UASNs. Node clustering is widely considered to be able to optimize network energy consumption. Although the current clustering-based routing method prolongs the network life cycle to a certain extent, some nodes may fail due to excessive energy consumption caused by excessive data transmission tasks, and the problems of high and uneven energy consumption still exist. To extend the life cycle of UASNs, this article proposes an unequal clustering method based on particle swarm optimization. Our method uses iterative updates to dynamically adjust the cluster size based on the remaining energy of the cluster head, the distance from the cluster head to the Sink node, and the number of times the cluster head forwards data between clusters to balance the cluster head load. The energy consumption of the whole path from the cluster head to the Sink node and the number of hops required are considered in the intercluster transmission phase. Simulation results show that this method can effectively reduce the network energy consumption.**

***Index Terms*—Clustering, energy consumption optimization, underwater acoustic sensor networks (UASNs).**

I. Introduction

APPROXIMATELY 71% of the Earth's surface is covered by seawater, and the ocean is rich in resources. Hence, underwater acoustic sensor networks (UASNs) are important for ocean information acquisition and are widely used in underwater resource exploration, data collection, and underwater environmental monitoring. The UASNs consist of a Sink and a number of acoustic sensor nodes deployed underwater. Acoustic sensor nodes are deployed randomly or exclusively in the area to be monitored, with functions of data acquisition, information sensing, data processing, and data transmission. After completing data acquisition, the data are forwarded to the Sink node through single-hop or multihop.

The underwater acoustic sensor node is usually powered by its own power supply or its own electrical energy reserve. Because its working environment is underwater, there are certain difficulties in power supply renewal or electrical energy replenishment, and the node will not work once it runs out of energy. Therefore, energy consumption optimization has become the focus of research on UASNs [1], [2].

The energy consumption of UASNs mainly comes from data transmission. Thus, optimizing the data transmission path can reduce the energy consumption of the network. Clustering is an effective energy-saving scheme. It classifies sensor nodes into two types: 1) cluster head nodes (CHs) and 2) non-cluster head nodes (nCHs) [3], [4], where the nCHs are responsible for collecting information and sending the data to the cluster head, and the CHs are responsible for collecting information within the cluster and sending the data to the Sink node in a single-hop or multihop manner. The cluster-based routing approach reduces the energy consumption burden of nCHs by shortening the transmission distance and reduces the data transmission volume by using CHs to fuse the data within clusters, which offers an energy-saving solution that can effectively extend the network life cycle.

In UASNs, the energy decay is affected by the transmission distance. If CHs communicate with the Sink node in a single-hop manner, nodes far away from Sink consume more energy to transmit data than nodes close to Sink, which can easily fail due to energy exhaustion, thus reducing the coverage of the network. However, in multihop transmission, the nodes closer to Sink may act as intercluster relays more frequently and consume more energy between clusters, leading to the "hotspot" problem, i.e., some nodes consume too much energy at too fast a rate due to excessive transmission tasks, and the imbalance of energy consumption affects the lifetime of the whole network. To solve the above problems, an unequal clustering method based on particle swarm optimization (PSO) algorithm is proposed, which can alleviate the "hotspot" problem by dynamically adjusting the cluster size and make the energy

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consumption more balanced. The main contributions of this article are as follows.

1) A PSO-based cluster head selection method is proposed to construct an adaptation value function based on network energy consumption and node residual energy, and then CHs are selected using PSO.
2) An intercluster transmission method is designed to determine the optimal transmission path from each CH to Sink based on the energy consumption and the required number of hops.
3) To balance the energy consumption among CHs, an unequal clustering method is proposed to effectively alleviate the "hotspot" problem of UASNs by dynamically adjusting the cluster size.
4) The proposed method is compared with existing methods in terms of single-round energy consumption, residual energy, and node death.

The structure of this article is as follows. Section II reviews the existing work, Section III introduces the structure of UASNs and the energy consumption model, Section IV presents the proposed UCPSO method in detail, Section V discusses and analyzes the experimental results, and Section VI concludes.

II. Related Work

To optimize the network energy consumption of UASNs, researchers have conducted considerable research on clustering algorithms. Heinzelman *et al.* [5] proposed the low energy adaptive clustering hierarchy (LEACH), which selects cluster heads in a cyclic manner, and all nodes have the opportunity to act as cluster heads within a certain number of rounds, and the energy load of the whole network is distributed to each sensor node as evenly as possible, thus effectively balancing the energy consumption. However, due to the randomness of cluster head selection, it is easy to lead to the uneven energy of the whole network.

Khan *et al.* [6] proposed a multilayer clustering energy-saving scheme, which first stratifies the whole network, selects the appropriate CHs based on the Bayesian probability and residual energy, and each CH selects the appropriate next hop based on the fitness value. Yu *et al.* [7] proposed an energy-optimized clustering scheme, which considered several factors in cluster head selection, including the number of neighboring nodes of each underwater acoustic sensor node, the current remaining energy, and the distance to the Sink node. To control the energy consumption, an energy optimization mechanism is designed by introducing an inverse tangent function, which adaptively adjusts the maximum effective communication distance of this node according to the ratio of the node's remaining energy to the initial energy, and the communication range of the node gradually becomes smaller as the energy consumption of the node increases, thus extending the node lifetime.

Wan *et al.* [8] proposed an adaptive clustering algorithm, where the selection of CH takes into account the residual energy and the distance between nodes. The selection of the next hop between clusters is based on the residual energy of the nodes and the designed routing rules, and a more energy-efficient single-hop or multihop approach is chosen, which reduces the energy loss during transmission.

Ahmed *et al.* [9] proposed a redundancy control-based clustering method to reduce energy consumption by removing redundant data before the region head and CH forward the data to the next layer.

Hou *et al.* [10] proposed an energy-balanced unequal layering clustering scheme (EULC). This scheme first divides the network into uneven layers based on depth, and the layer spacing is wider for layers farther away from the Sink, while the competition radius of the nodes is adjusted according to the distance from the nodes to the Sink node, making the clusters closer to the Sink node smaller, which effectively alleviates the "hotspot" problem. In the process of cluster head selection, the influence of node residual energy, distance to Sink and node degree is fully considered, and each node calculates the competition power value to determine the set of candidate cluster heads, in which the node with the largest competition power value in the candidate cluster headset is elected as the official cluster head.

Wadud *et al.* [11] proposed an energy-balanced reliable routing protocol, which includes a routing mechanism based on node depth that balances the energy consumption among neighboring nodes by the remaining energy of the next-hop forwarding node and the number of potential next-hop forwarding nodes. The nodes adjust their own transmission power adaptively according to the distance of the farthest node in the list of neighboring nodes, which improves energy efficiency. Wang *et al.* [12] proposed a clustering algorithm based on node density and a Voronoi diagram, which selects cluster heads based on node density. The network is divided into multiple circular regions, and the cluster heads are located in the center of the regions. A node auto-adjustment algorithm based on a Voronoi diagram is proposed to achieve energy balance, which makes the number of member nodes in each cluster head nearly the same, effectively alleviating the problem of the uneven distribution of cluster heads in the network. Zhao *et al.* [13] proposed a dynamic hierarchical clustering method for balancing the energy consumption of 3-D WSNs networks, which introduced a new 3-D sphere model combined with an energy consumption model to determine the optimal number of clusters in the network, and introduced a distance similarity index to improve the clustering performance, and proposed a dual-cluster head strategy for the problem of high cluster head load of large clusters. In addition, the method constructs the optimal cluster head function based on the residual energy and location information of nodes to determine the optimal cluster head selection result for each cluster.

Lin and Wang [14] proposed a game theory-based energy-efficient clustering algorithm, which mitigates the energy overhead of cluster heads by using a dual-cluster head strategy while proposing a noncooperative game model to achieve energy balance among cluster heads. Wang and Tong [15] proposed an ant colony algorithm-based clustering method, which improves the LEACH by adding the influence of residual energy and node location in the cluster head selection threshold function of LEACH. Thus, nodes with high residual

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energy have a higher probability of being selected as cluster heads, whereas nodes at the edges of the network have a lower probability of being selected. In the intercluster transmission phase, the best data transmission path of nodes is selected iteratively using the ant colony algorithm.

Wang *et al.* [16] proposed an artificial bee colony algorithm-based clustering scheme, where the initial energy of all nodes is the same when the network is not working in the initial state, combined with the K-means algorithm to establish the initial clusters. When the energy of the nodes in the network is heterogeneous, a modified artificial bee colony algorithm is used to select the cluster heads. The residual energy, location, and density of the cluster heads jointly affect the cluster head selection results. In addition, an energy-efficient intercluster routing scheme based on ant colony algorithm optimization is proposed to determine the optimal data transmission path from all cluster heads to the Sink node. Daneshvar *et al.* [17] proposed a routing method for WSNs based on the gray wolf optimization algorithm, which predicts the possible energy consumption of nodes in the actual network, constructs an adaptation value function by combining the remaining energy of the nodes, and then selects cluster heads based on several iterations of the gray wolf optimization algorithm.

Pachlor and Shrimankar [18] proposed a cluster partitioning strategy for addressing the overload that occurs when all nCHs in a cluster are unable to act as cluster heads. When the energy level of the nodes in the cluster is low, then the cluster is divided into two subclusters to reduce the energy consumption of the cluster heads, saving the energy used to re-establish the cluster. Liu *et al.* [19] proposed a topology control strategy based on a complex network theory, which uses a scale-free network model to generate the initial topology and a dual cluster head structure with two types of cluster heads used to maintain the connectivity as well as the coverage of the network. Song *et al.* [20] proposed an intelligent collaborative balancing scheme for enhancing network service quality and resisting malicious intrusions. Behera *et al.* [21] proposed an energy-efficient cluster head selection scheme, which improves the threshold function of the LEACH algorithm by introducing two influencing factors, the ratio of the remaining energy to the initial energy and the derived optimal value of the number of cluster heads, to effectively balance the network load.

In recent years, many researchers have applied the PSO algorithm to UASNs. Su *et al.* [22] proposed a PSO-based cluster head selection method, introduced the variational crossover operator of a genetic algorithm to improve PSO, accelerated the convergence speed, and selected cluster heads with load balancing and energy consumption as optimization objectives. The life cycle of the network is thereby effectively extended. Zhang *et al.* [23] proposed a globally optimized clustering algorithm, which uses a discrete PSO algorithm and genetic algorithm for CH selection, and the design of the adaptation value function considers the residual energy of the cluster head, the load of the cluster head, and the range of the cluster, which effectively reduces the network energy consumption. Li *et al.* [24] improved the previous work [23] and proposed an improved PSO algorithm, which shows that the required node transmission power and transmission distance have a nonlinear relationship, and the optimization problem should focus on the transmission power rather than the transmission distance. Wang *et al.* [25] proposed an algorithm to search for energy centers using the PSO algorithm, which selects the nodes near the energy centers as CHs to alleviate the "hotspot" problem. However, the algorithm only considers the influence of energy in selecting CHs and ignores the influence of the distance between nodes.

Sahoo *et al.* [26] proposed a PSO-based cluster head selection optimization method, which takes into account five influencing factors: 1) residual energy of nodes; 2) distance; 3) node degree; 4) energy consumption rate; and 5) average energy, to derive the best cluster head selection results. Kaur and Grewal [27] proposed a PSO-based dual movable Sink node clustering scheme, using the PSO algorithm to optimize the selection of cluster heads, where the result of cluster head selection will be determined by the ratio of remaining energy to initial energy, node degree, node centrality, distance between nodes and Sink, number of cluster heads, and energy consumption rate together.

Krishnaswamy and Manvi [28] proposed a clustering method based on fuzzy clustering and PSO. A fuzzy clustering algorithm is used to divide the nodes into clusters, and then a PSO algorithm is used to select the cluster head from each cluster. The objective function considers the intracluster transmission distance, the distance from the cluster head to the Sink, and the energy consumption. Fei *et al.* [29] proposed a hybrid clustering method combining a fuzzy clustering algorithm and moth flame optimization method inspired by [28], which first uses a fuzzy clustering algorithm to divide the nodes into clusters and then selects the optimal CH in each cluster using the moth-flame optimization method.

The current applications of the PSO algorithm in UASNs are mostly based on the idea of uniform clustering, which ignores the intercluster transmission of data and the fact that the nodes far from the Sink often need to consume more energy and the resulting imbalance of energy consumption shortens the life cycle of the network. It has been shown that the unequal clustering approach can effectively balance the energy consumption of cluster heads. If a cluster head consumes more energy for intercluster transmission, the communication overhead of that cluster head within the cluster is moderately reduced to save energy to achieve energy balance. In this article, we propose a PSO-based unequal clustering method for hydroacoustic sensor networks using the PSO algorithm to select cluster heads. We then balance the CH load by controlling the cluster size to even the energy consumption within and between clusters. The experimental results show that the method can effectively balance and reduce the energy consumption of the whole network and extend the network life cycle.

III. System Model

A. Network Model

This article targets 3-D UASNs, as shown in Fig. 1, where a Sink node is deployed as surface receivers in the center of the surface above the cubic region and sensor nodes are

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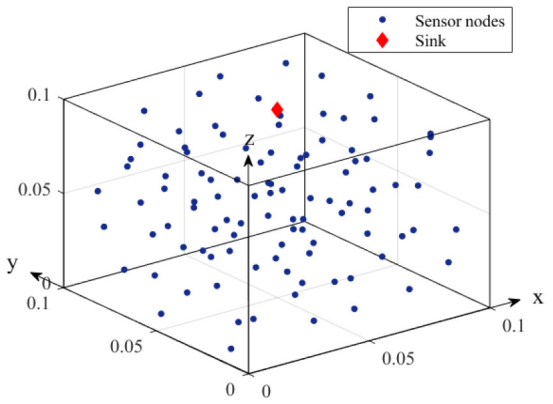


Fig. 1. Structure of UASNs.

distributed throughout the region in a random deployment. To facilitate the experiments, the following assumptions are made about the network as well as the sensor nodes.

1) The Sink node location is fixed, its energy is unlimited, and the Sink node knows the location information of all nodes.
2) All nodes are uniquely numbered, have the same initial energy, and can control their own transmission power.
3) The communication link between nodes is reliable.
4) All sensor nodes send the collected data periodically.

B. UASN Energy Consumption Model

The communication energy consumption model in this article relies on [30], where the energy consumed by a node forwarding l bits of data to a node at distance d is calculated as follows:

$$\text{Esend}(l, d) = P_0 * A(d) * T \tag{1}$$

where P_0 is the minimum power required by the node to receive the data, T denotes the transmission time of the packet, and $A(d)$ denotes the energy decay at a transmission distance of d. The calculation of $A(d)$ is as follows:

$$A(d) = \mathrm{d}^k \cdot a^d \tag{2}$$

where k is the energy diffusion factor and a denotes the attenuation coefficient, calculated by

$$a = 10^{\alpha(f)/10} \tag{3}$$

$\alpha(f)$ is the absorption factor, which can be expressed as follows:

$$\alpha(f) = \frac{0.11f^2}{1+f^2} + \frac{44f^2}{4100+f^2} + \frac{2.75f^2}{10000} + 0.003 \tag{4}$$

where f is the signal frequency. During data transmission underwater, the energy attenuation is mainly affected by the distance as well as the signal frequency, which limits the underwater long-distance communication and the maximum bandwidth utilization.

The energy consumption of the node to receive l bits of data is $E_R(l) = l \cdot E_{\text{elec}}$, where E_{elec} is the energy consumption of the node to process one bit of data. The energy consumption of the node to fuse l bits of data is $E_I(l) = l \cdot E_{\text{integrate}}$, where $E_{\text{integrate}}$ is the energy consumption of the node to integrate one bit of data.

IV. UCPSO Method

A. CH Selection

A PSO-based cluster head selection method is proposed for the energy consumption characteristics of the underwater environment.

1) Particle Encoding: A particle is a possible CH selection result where each particle is a sequence of numbers consisting of "0" or "1," with each bit in the sequence corresponding to a surviving node in the network. The number "1" in the sequence means that the corresponding node is selected as CH, and the number "0" indicates that the corresponding node is nCH.

2) Initialization: Since the energy consumption of CH nodes is much larger than that of nCH nodes, the nodes with higher residual energy need to be selected as CHs as much as possible. To protect the nodes with low residual energy and to ensure a sufficient number of nodes to participate in CH selection, all surviving nodes are divided into candidate nodes and normal nodes according to the set candidate node ratio, where the candidate nodes are the nodes with relatively higher residuals. To make the CH distribution more uniform, the interval threshold R_0 is set between CHs, and the particle initialization as well as the particle update must check whether the CHs selected by that particle satisfy the interval condition.

During the initialization process, the particle population $PS = \{PS_1, PS_2, \ldots, PS_i, \ldots\}$ is randomly generated according to the CH election ratio P_{CH}.

3) Fitness Function: The fitness function is used to judge the performance of the solution, and each solution represents a cluster head selection result. A reasonable clustering result should ensure that the residual energy of the CHs is as high as possible, while the total energy consumption of a round is as low as possible. Therefore, the residual energy and total energy consumption of the cluster head are considered as the optimization objectives in the design of the fitness function.

Due to the high energy consumption of the CHs, nodes with more sufficient residual energy should be selected to participate in cluster head selection. The first objective optimization function is $f_1 = E_0/\overline{E_{\text{CH}}}$, where $\overline{E_{\text{CH}}}$ is the average value of the residual energy of all nodes whose current particles are marked as cluster heads, and E_0 is the initial energy value of a single node. The higher the average residual energy of the cluster head, the smaller the value of the objective function f_1.

The total energy consumption of a round is incorporated into the evaluation of the particles, and the CHs and nCHs selected by the current particle simulate a round of data transmission according to the proposed cluster formation method as well as the intercluster transmission method to derive the predicted total energy consumption E_{total} for a round. The second objective optimization function is $f_2 = E_{\text{total}}$.

Since the values of f_1 and f_2 are of different magnitudes, the values are mapped to between 0 and 1 after processing using the Sigmoid function $S(x) = (1/1 + e^{-x})$ and then compared. Then the multiobjective is weighted to make

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it a single-objective optimization, resulting in an adaptation value function of fitness $= m \cdot f_1 + n \cdot f_2$ with $m+n=1$, where m and n are weighting factors that add up to 1. When f_1 is smaller then the average residual energy of the cluster head is higher, while f_2 is smaller then the energy consumption is lower, therefore, the smaller the fitness value of the particle fitness, the better the solution is indicated.

4) Particle Updating: After initialization, the fitness values of all particles are calculated and compared to obtain the historical optimal solution set HB of all particles, $HB = \{HB_1, HB_2, \ldots, HB_i, \ldots\}$ and the global optimal particle GB.

The particle is updated by tracking the current historical optimum as well as the global optimum. Particle PS_i is first crossed with the historical optimum HB_i of the particle with a certain probability p_1. Then two CHs of the two groups of particles are randomly selected, the current particle PS_i is updated after swapping, and the operation is repeated if there is a duplicate CH or the CH interval condition is not satisfied. After crossover, the fitness value of the updated PS_i is calculated, and HB_i and GB are updated.

Similarly, the updated PS_i is crossed with the global optimal particle GB with probability p_2. To speed up the convergence, a mutation operation is performed on the updated PS_i with probability p_3, i.e., a node is randomly selected in the candidate node set to replace a certain CH of the current particle PS_i. When duplication and failure to satisfy the CH interval occur, the operation is repeated. After performing the mutation operation, PS_i, HB_i, and GB are updated.

After reaching the maximum number of iterations, GB is taken as the optimal CH selection result. CH selection is performed by the Sink node because the Sink node can replenish the energy and has more processing power and higher storage capacity. The Sink node broadcasts the result after the CH selection is completed. The CH selection process is shown in Algorithm 1.

Algorithm 1 Proposed Method to Select the CHs

```
Initial PS
Calculate the fitness value of all particles
HB ← PS
GB ← PS_i whose fitness value is minimum
for iteration ← 1 to Max_iteration
  for each particle PS_i do
    if rand < P_1
      PS_i ← PS_i crossing with HB_i
    end if
    update HB_i, GB
    if rand < P_2
      PS_i ← PS_i crossing with GB
    end if
    update HB_i, GB
    if rand < P_3
      PS_i ← PS_i after variation
    end if
    update HB_i, GB
  end for
end for
```

B. Intercluster Transmission

After the set of selected CHs is known, the next hop of each CH between clusters is selected globally by the Sink node in order to achieve an optimal selection of intercluster messaging, which is implemented as follows.

Step 1: All CHs are sorted according to their depths in order of proximity to the water surface, and finally the set US of initial ordered CHs is obtained, and US denotes the set of cluster heads that have not selected the next hop.

Step 2: Establish the empty set PN, PN indicates the set of nodes that may be used as the next hop, and then add the Sink node to the empty set PN, the first node of the PN set is Sink, indicating that the Sink node can be used as the next hop of the CH between clusters and also as the end of the data transmission path.

Step 3: The nodes in the US set select the next hop from the set PN in turn, remove the node that has selected the next hop from the US set, and then add the node to the PN set. It is guaranteed that all CHs can find the appropriate next hop in the order of depth from smallest to largest, and the transmission path from each CH to Sink is planned.

Each CH in US finds the best next hop node in the PN set. The cost function of a node in the PN as the next hop of the CH is as follows:

$$W = E \cdot 0.8^P \tag{5}$$

where E denotes the total energy consumption on the path for a cluster head to send 1bit data after taking the node in PN as the next hop, and the path starts from this CH and ends at the Sink node. P denotes the number of hops required from Sink node after taking the node in PN as the next hop. Each CH selects the node in the PN with a smaller cost Was the next hop until the US set is empty, which means that the data transmission path from all CHs to Sink has been established.

Step 4: After determining the next hops of all CHs, the number of times $RN$of each CH involved in intercluster relay forwarding in a round of data transmission is counted and used to regulate the cluster size.

C. Unequal Clustering

The distribution of nodes and the amount of residual energy affect the energy consumption of nodes. If this is not taken into account, it will lead to large differences in energy consumption of the whole network and affect the network lifetime. Hence, the cluster size needs to be controlled according to the specific situation to balance the differences in energy consumption among CHs.

In this article, a preset range r is defined for each CH in order to control the size of the CH into clusters. The range r is influenced by the remaining energy of the CH, the distance to the Sink node and the participation in intercluster relaying and is given by the following equation:

$$r = \sqrt[3]{R_0^3 \cdot \left(\alpha \frac{E_{CH_i} - \overline{E_{\mathrm{CH}}}}{\Delta E} + 1\right) \cdot 0.6^{RN} \cdot \left(\beta \frac{d_{\mathrm{to_sink}}}{d_{\max}} + 1\right)} \tag{6}$$

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where R_0 is the initial distance, E_{CH_i} denotes the residual energy of the ith CH, $\overline{E_{\mathrm{CH}}}$ is the average value of the residual energy of a set of CHs, and ΔE is the set energy difference (set to 0.003, which is an approximation of the average energy consumption of CH nodes after one round of data transmission), which is used to evaluate the size of the difference between the residual energy of CHs and the average value. RN is the number of times CH_i participates in relaying during intercluster transmission, $d_{\mathrm{to_sink}}$ denotes the distance from CH_i to the Sink node, $d_{\max}$ is the maximum distance of the nodes in the network from the Sink, and the parameters α and β denote the degree of influence of residual energy and distance to the Sink on the range r, respectively. The higher the residual energy, the larger the r. The larger the distance to the Sink node, the larger the r. If the CH node is involved in intercluster relaying, r decreases moderately. Usually, CH nodes near the Sink node are more likely to die due to relaying a large amount of data, leading to a "hotspot" problem. The proposed method takes into account the CH properties and CH participation in intercluster transmission as much as possible, which can better balance the energy consumption among CHs and alleviate the "hotspot" problem of the network.

Sink node also participate in the process of clustering, i.e., nCH nodes close to the Sink node can transmit data directly to Sink without establishing a link with a particular CH, which further relieves the relay pressure on CH nodes relatively close to Sink. the preset range r of the Sink node takes the initial value R_0.

All nCH nodes calculate the cost w of clustering with each CH based on the distance as well as the range r. The CH with the smallest clustering cost w is selected to cluster or transmit data directly to the Sink node, and the clustering cost w is calculated as follows:

$$w = d_{\mathrm{to_CH}_i}/r \tag{7}$$

where $\mathrm{d}_{\mathrm{to_CH}_i}$ is the distance from the nCH to the CH or Sink node and r is the predefined range of the CH or Sink. The clustering of nCH nodes is influenced by the distance and clustering with close CH or the Sink nodes as much as possible, and also constrained by the CH preset range r, so that the size of the whole network cluster presents a nonuniform state, thus achieving the purpose of balancing the energy consumption among CHs.

The time complexity can be used to describe the running time of an algorithm. The proposed method UCPSO is carried out in rounds, so the time complexity of each round of cluster head selection is analyzed, assuming that the number of particles in the round is N, the maximum number of iterations is m, and each iteration time is T, thus the time required for cluster head selection is $N \times m \times T$. Each iteration time T mainly comes from the time consumption of calculating the adaptation value before particle update. Assuming that the number of sensor nodes in the network is n, then the time complexity of T is $O(n^3)$.

V. Simulation Results

To verify the effectiveness of the proposed method, MATLAB simulation experiments were used for comparison with LEACH [5], EULC [10], and IPSO [24]. The main simulation parameters are shown in Table I.

TABLE I
Simulation Parameters

Parameter	Value
Network Scope	100 m*100 m*100 m
Number of nodes	100
Location of sink	（50 m,50 m,100 m）
Packet size	2000bit
Initial energy	0.1J
P_0	3mW
f	10 kHz
E_{elec}	50nJ/bit
$E_{integrate}$	5nJ/bit
R_0	20 m
P_{CH}	0.1
Particle swarm size	100
Breeding Algebra	50

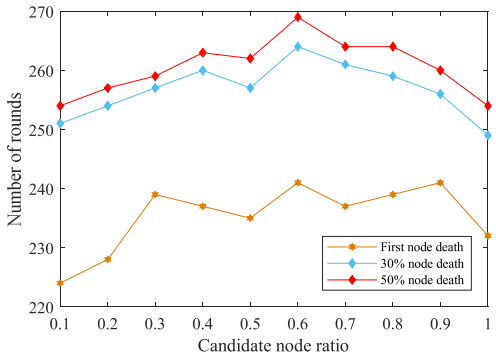


Fig. 2. Effect of candidate node ratio on node death.

The performance of UASNs is evaluated in terms of both network energy consumption and network life cycle. Network energy consumption is an important performance evaluation metric in network routing design, which is usually the sum of the energy consumption of the network, i.e., the total energy consumption of all sensor nodes. The lower the network energy consumption is, the higher the energy utilization is. The network life cycle is the time period from when the network is deployed to start working underwater to when the network fails completely, in "rounds", where the time span of a round starts when all nodes in the network start collecting data and ends when the Sink node receives the data transmitted by all nodes. Different application scenarios may have different performance requirements for the network, and the definition of the network life cycle also changes, with the common definitions of first node death time, 30% node death time, 50% node death time, and all node death time. In this article, we analyze the first, 30%, and 50% node death cases.

Fig. 2 shows the effect of different candidate node ratios on the number of node death rounds in the UCPSO method. Candidate nodes are determined based on the residual energy of the nodes, and it can be found that the best results are achieved when the candidate node ratio is set to 60%. Candidate nodes, as nodes eligible to be elected as cluster heads, are ranked by the residual energy of all surviving nodes in the network during the initialization phase, and then

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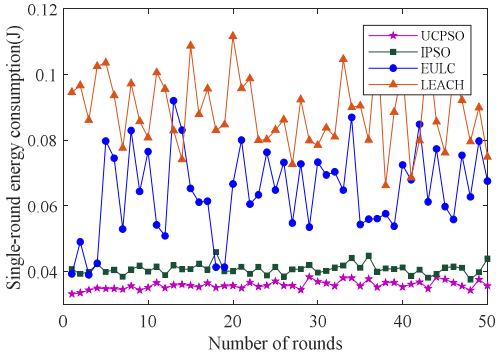


Fig. 3. First 50 rounds of single-round energy consumption comparison.

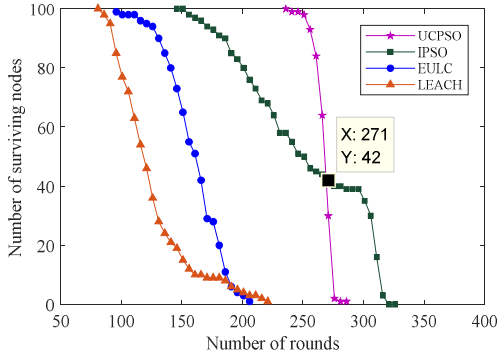


Fig. 4. Comparison of the number of surviving nodes.

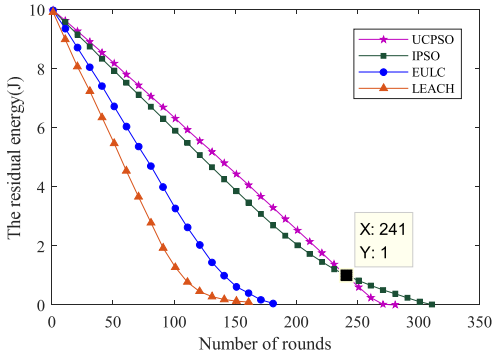


Fig. 5. Residual energy comparison.

the set of candidate cluster heads is determined according to the set candidate node ratio, which is used to determine the number of candidate nodes, so the candidate nodes have a relatively higher residual energy. If the number of candidate nodes involved in the iteration is not enough or too many, it will affect the result of cluster head selection. The higher the number of candidate nodes, the higher the probability that the nodes with lower residual energy will be elected as cluster heads, which speeds up the death of low-energy nodes; the lower the number of candidate nodes, the lower the number of nodes involved in the iteration, which affects the performance of the cluster head selection result.

Fig. 3 shows the total energy consumption per round for the first 50 rounds of the four methods. It can be seen that LEACH has the highest single-round energy consumption and is not sufficiently stable. This is due to the random selection of CHs by LEACH and the unstable number of cluster heads, resulting in large energy consumption fluctuations. EULC reduces the single-round energy consumption by considering node properties, such as residual energy, node degree, and distance to the Sink node. IPSO uses the PSO algorithm for optimization while considering the residual energy of nodes, transmission power required by nodes, and cluster head load, which greatly reduces the energy consumption per round. UCPSO, while fully considering various attributes of nodes, adjusts the corresponding cluster size according to the participation of CHs in intercluster relaying. In addition, compared with IPSO, CH transmits data to the Sink node in a multihop or single-hop manner, and nodes close to Sink can directly interact with Sink nodes in a more reasonable way for intercluster transmission. The energy consumption and fluctuations of UCPSO are minimal in a single round, which means that the energy consumption is smaller and more balanced in this method.

Fig. 4 shows the changes in the surviving nodes of the whole network as the number of simulation rounds increases. The proposed method significantly improves the node survival rate compared with LEACH, EULC, and IPSO. Moreover, it delays the death of the first node, avoids the problem of insufficient network coverage due to the premature death of nodes, and improves the energy utilization. LEACH relies on the comparison of random numbers and threshold value to determine whether to be a candidate node, the number of cluster heads is variable and unevenly distributed, some nodes may die prematurely due to frequent election of cluster heads, and all cluster heads have been communicated with the Sink node in the form of a single hop, the energy consumption of nodes far from Sink is still high, therefore, the number of surviving nodes decreases sharply. EULC takes into account the influence of factors, such as residual energy of nodes and distance from the Sink node, and intercluster transmission nodes forward data in single-hop or multihop form, which effectively reduces network energy consumption. IPSO utilizes a PSO algorithm for optimization, which determines the objective optimization function based on residual energy, transmission power required by nodes, and load of cluster heads, making the selected cluster heads more reasonable and effectively extending the network life cycle. The number of surviving nodes decreases sharply with the increase of rounds after the death of the first node in the proposed method UCPSO, and after running up to 271 rounds, the number of surviving nodes in the proposed algorithm is lower than that of IPSO. Since IPSO is based on uniform clustering and the cluster head communicates directly with Sink in a single-hop manner, the nodes far from Sink will die faster, and the nodes near Sink have lower energy consumption and survival time. Nodes nearer to Sink have lower energy consumption survive longer and have unbalanced energy consumption. And the proposed algorithm delays the death of nodes, compared to IPSO, the network has a wider coverage, which leads to more energy consumption, so the number of surviving nodes decreases rapidly at a later stage.

Fig. 5 shows the change in the total residual energy of the whole network during the simulation experiment. UCPSO dynamically adjusts the cluster size by nonuniform cluster

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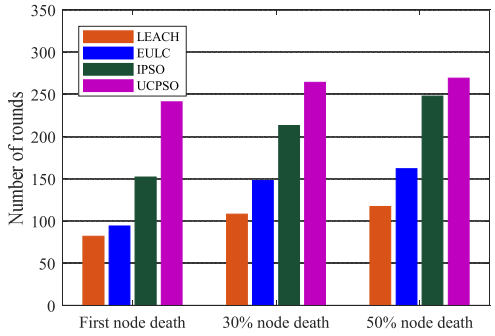


Fig. 6. First, 30%, 50% node death rounds comparison.

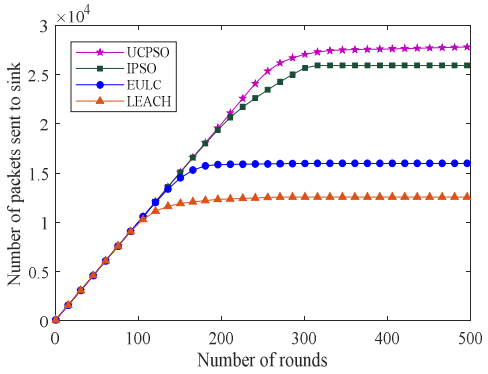


Fig. 7. Comparison of the number of packets sent to Sink.

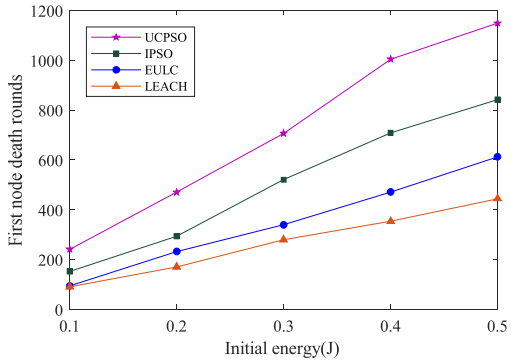


Fig. 8. Comparison of first node death rounds with different initial energy.

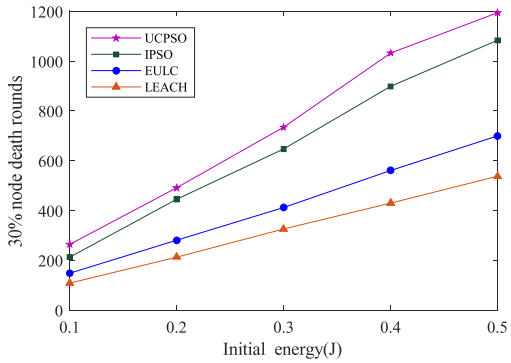


Fig. 9. Comparison of 30% node death rounds with different initial energy.

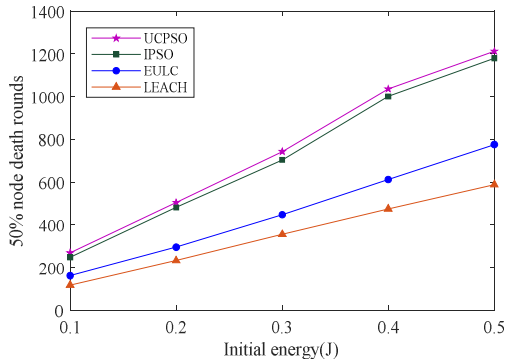


Fig. 10. Comparison of 50% node death rounds with different initial energies.

splitting, which makes the selection of cluster heads more rationalized, and intercluster communication in a single-hop or multihop manner further reduces the network energy consumption, so the decrease in residual energy is the lowest. However, the residual energy of the two methods involving PSO crosses when running up to 241 rounds, after which the residual energy of IPSO is higher, which is due to the higher number of surviving nodes of the UCPSO method in the early stage, resulting in higher energy consumption and relatively lower total residual energy of the network by the later stage. From Fig. 4, it can be seen that the number of surviving nodes is the same for both UCPSO and IPSO methods at round 271. At 241 rounds, the number of UCPSO surviving nodes is much higher than IPSO, and at this time both methods have the same residual energy, indicating that the proposed method has a higher utilization of UCPSO energy.

Fig. 6 shows the comparison of the first node death, 30% node death and 50% node death rounds, and it can be found that the proposed method greatly delays the death of nodes and extends the lifetime of the network.

Fig. 7 compares the changes in the number of packets sent to the Sink node as the simulation time increases. All nodes in UASNS need to send the collected data to the Sink node, so the number of packets received by the Sink node can effectively evaluate the working performance of the network. It can be seen from the figure that the number of packets received by the four methods is the same when there is no death of nodes, and as the number of simulation rounds increases, the total number of packets received by the UCPSO method is more than the other three methods, which is because the nonuniform clustering approach in this article extends the working time of sensor nodes and the energy consumption of the whole network is more balanced, so the proposed UCPSO method Sink node has a higher packet reception rate is higher.

Figs. 8–10 show the comparison of the first node death rounds, 30% node death rounds and 50% node death rounds of the four methods for sensor nodes under different initial energy conditions, respectively. It can be seen that the node survival rate of the proposed method is always better than the other three under the same energy conditions. Consequently, the proposed method extends the life cycle of the network more effectively.

Fig. 11 shows the average energy consumption per round for the first 50 rounds of the four methods when the number of nodes in the network varies. The network does not experience node death in the first 50 rounds, so the average

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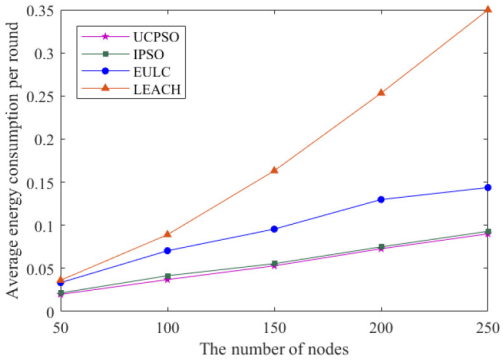


Fig. 11. Comparison of the average energy consumption per round for a different number of nodes.

single-round energy consumption of the first 50 rounds is taken as the performance evaluation index to assess the impact of the number of nodes on the performance of the proposed method. It can be found that the proposed method still outperforms the other three methods in reducing the energy consumption of the network when the number of nodes varies.

VI. Conclusion

In this article, a PSO-based unequal clustering method for UASNs is proposed to address the problem of the short life cycle of UASNs. The residual energy and distance constraints of nodes are considered in cluster head selection, and in addition, the influence of cluster heads on intercluster transmission is taken into account to unequally cluster the whole network. This makes the cluster head selection results more reasonable. The simulation results show that the method can effectively balance the energy consumption differences in the network and extend the network life cycle. However, since the sensor nodes are deployed underwater, they are easily affected by the environment, leading to changes in the network topology. In this light, we note that this article only considers the impact of energy consumption on the network without considering the connectivity of the network and the reliability of information transmission.

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